

Yash Thacker

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PROFESSIONAL PROFILE

- PhD student in Imaging Science with experience building scalable medical-imaging software, HPC pipelines, and machine-learning workflows for neuroimaging, fNIRS/DOT, radiomics, and clinical image analysis.
- Strengths include independent problem solving, cross-institutional collaboration, reproducible software development, statistical validation, and translating complex imaging methods into tools usable by researchers and clinicians.

EDUCATION

Washington University in St. Louis PhD Imaging Science + M.S. Biomedical Engineering	Aug 2024 - Current
• GPA: 3.95; research focus: machine learning applications to medical imaging, neuroimaging, and diagnostics.	
Washington University in St. Louis B.S. Computer Science; Minor in Bioinformatics	May 2020
• GPA: 3.77; Honors: Harold P. Brown Engineering Fellowship, Cum Laude, APIDA Honor.	
Illinois Wesleyan University B.S. Biology; Minor in Mathematics	Dec 2018
• GPA: 3.92; Honors: Summa Cum Laude, Phi Kappa Phi, Tri-Beta, NSLS, Mortar Board.	

RESEARCH EXPERIENCE

Medical AI Lab Washington University in St. Louis PhD Student	Jan 2025 - Current
• Developed two lab-wide CHPC neuroimaging pipelines, mapperTrac_CHPC and MATLAB_TBM_CHPC, supporting diffusion MRI preprocessing, tractography, connectome generation, transport-based morphometry, Jacobian export, QC, and downstream ML analyses. • Adapted MaPPeRTrac for WashU SLURM-based computing by building BIDS-compatible subject/session/run discovery, DICOM-to-BIDS automation, denoising, Gibbs correction, eddy correction, eddy QC, T1 QC, validation, and rerun logic. • Converted TBM workflows into scalable batch-processing infrastructure for subject registration to the HCP template, deformation-field estimation, parameter grid search, per-subject PNG QC summaries, and NiftI Jacobian maps for voxel-wise statistics and ML. • Built monitoring dashboards and status-tracking tools to identify completed jobs, failed runs, convergence issues, numerical instability, and robust parameter sets across large neuroimaging cohorts. • Developed statistical and ML workflows for 16p11.2 CNV neuroimaging, including GLM regression, Benjamini-Hochberg FDR correction, age/sex/brain-volume confound regression, PCA, regularized LDA, Random Forest models, and cross-validation evaluation. • Worked closely with UCSF collaborators to validate pipeline assumptions and connect TBM-derived morphometry with diffusion MRI measures of white-matter microstructure. • Mentored undergraduate research assistants on experimental design, machine learning methods and results interpretation; facilitated successful independent project milestones and data analysis.	
Brain Light Laboratory Washington University in St. Louis PhD Rotational Student	Sep 2024 - Dec 2024
• Developed a real-time fNIRS processing pipeline for potential hospital patient-monitoring applications, including ICU and ECMO populations at elevated neurological risk. • Reduced NeuroDOT data processing time from approximately 20 seconds to 0.2 seconds per 120 frames and optimized visual-cortex mesh display from 20 seconds to 9 seconds. • Analyzed NeuroDOT workflows to identify essential components, optimization opportunities, and steps suitable for redesign or removal.	
Neuroinformatics Research Group Washington University in St. Louis Research Developer II	Jul 2023 - Aug 2024
• Led development of a new XNAT fNIRS data type in Java, including zip-data importing, assessor/scan/session/subject components, and integration with legacy XNAT code. • Deployed the fNIRS plugin for OXI, an open-source cloud-based platform supporting multi-institutional collaboration and a \$5M grant-funded project. • Integrated Jupyter notebook-based analysis into XNAT so researchers could run workflows and view pipeline outputs directly in the browser. • Co-led NeuroDOT and OXI workshops at OHBM and fNIRS conferences; instructed approximately 30 novice-to-advanced users on containers and analysis workflows.	
Neuroinformatics Research Group Washington University in St. Louis Research Developer I	Sep 2020 - Jun 2023
• Developed Dockerized XNAT pipelines for DICOM filtering, preprocessing, CNN training, hyperparameter grid search, and heatmap-based interpretation for COVID-19 chest radiograph classification. • Created a pyradiomics pipeline that rescaled OHIF ROI annotations, extracted texture features, and aggregated analysis-ready CSV outputs for clinical renal-lesion characterization. • Collaborated with the Brain Light Lab to co-author NeuroDOT Python, adapting a MATLAB toolbox into a Python-based connectomics analysis library for optical brain mapping.	
llumina Inc. San Diego, CA (Remote) Software Quality Engineer Intern	May 2020 - Aug 2020
• Co-authored a work instruction that streamlined script lifecycle compliance for low-risk script development and created an example script template for implementation. • Collaborated with stakeholders to evaluate metrics, improve script-development processes, and present final deliverables to senior quality leadership.	

TEACHING AND LEADERSHIP EXPERIENCE

AGES (Association of Graduate Engineering Students) | Social Chair

Aug 2025 - Current

- Planned and organized large-scale social events, including the graduate formal and networking happy hours, to foster community among 1000+ engineering graduate students.

ESE 4031 Optimization | Washington University in St. Louis | Teaching Assistant

Aug 2025 - Dec 2025

- Held weekly office hours and recitation-style support for an undergraduate optimization course covering linear/integer programming, simplex methods, mathematical modeling, and related computational problem-solving techniques.
- Graded assignments, helped students prepare for exams, and independently worked through homework problems in advance to support students without provided solution keys; invited to return for Fall 2026.

Math Crash Course | Washington University in St. Louis | Instructor and Course Designer

Jan 2025 – Aug 2025

- Collaborated with faculty and students to update course structure and develop linear algebra and calculus material for incoming students; designed homework and tutored students in twice-weekly sessions.

SELECTED RESEARCH SOFTWARE AND PROJECTS

mapperTrac_CHPC | Lab-specific MaPPeRTrac pipeline | [GitHub](#) | Python, Bash, SLURM, BIDS, FSL, FreeSurfer

- Built a WashU HPC-compatible DWI/T1 workflow for preprocessing, MapperTrac step sequencing, tractography, connectome generation, QC, dry-run testing, progress tracking, and rerun logic.

MATLAB_TBM_CHPC | Transport-Based Morphometry CHPC pipeline | [GitHub](#) | MATLAB, Bash, SLURM, Python

- Developed batch scripts and analysis workflows for subject to template registration, deformation-field estimation, Jacobian NIfTI export, parameter grid search, QC visualization, and ML-ready feature extraction.

fNIRS Docker Pipelines | NeuroDOT preprocessing, full processing, reconstruction | [Preprocess](#), [Full-Process](#), [Reconstruction](#) | Python, Docker, Bash

- Implemented NeuroDOT processing pipelines as Docker containers for XNAT to enable reproducible batch processing of fNIRS data.

fNIRS Plugin | XNAT data type and importer | [GitHub](#) | Java, JavaScript, HTML, XML

- Created an fNIRS data type for XNAT, including data importer, assessment declaration, and browser-accessible pipeline output views.

NeuroDOT Python | neurodotPY | [GitHub](#) | Python

- Adapted a MATLAB optical brain mapping toolbox into Python to improve accessibility for researchers and support broader use of NeuroDOT workflows.

Model Training Pipeline | DICOM finder, training, grid search | [DICOM Finder](#), [Training](#), [Grid Search](#) | Python, Docker

- Built containerized XNAT pipelines to filter DICOM scans, preprocess imaging data, train CNN models, and perform hyperparameter optimization.

Pyradiomics Pipeline | ROI resizer, feature extraction, collector | [ROI Resizer](#), [Pyradiomics](#), [Collector](#) | Python, Docker

- Created a clinical imaging pipeline to rescale ROI annotations, run pyradiomics feature extraction, and aggregate outputs for renal-lesion analysis.

PUBLICATIONS AND PRESENTATIONS

- Thacker Y**, Kundu S, et al. Differential White Matter Integrity Along the Corpus Callosum in 16p11.2 Deletion and Duplication Carriers. ASNR Scientific Abstract Power Pitches (Oral), May 2026.
- Thacker Y**, Kundu S, et al. Investigating Neural Pathways in Genetically-Mediated Autism using Generative Machine Learning. ASNR Showcase (ePoster + Oral), May 2026.
- Speh E, Segel A, **Thacker Y**, et al. Standardized assessment of fNIRS data quality with NeuroDOT and OXI. Diffuse Optical Spectroscopy and Imaging, Dec 2025.
- Speh E, **Thacker Y**, et al. NeuroDOT: a MATLAB and Python toolbox for diffuse optical tomographic brain mapping. Optical Tomography and Spectroscopy of Tissue XVI, Mar 2025.
- Speh E, Segel A, **Thacker Y**, et al. Standardized Assessment of fNIRS Data Quality with NeuroDOT and OXI. European Conference on Biomedical Optics, Jan 2025.
- Thacker Y**, Speh E, et al. Optical-imaging XNAT-enabled Informatics: an open-source platform for fNIRS data organization and processing. sFNIRS London (Poster), Sep 2024.
- Speh E, **Thacker Y**, et al. NeuroDOT: a MATLAB and Python-based self-contained toolbox for DOT. Neural Imaging and Sensing, Mar 2024.
- Speh E, **Thacker Y**, et al. NeuroDOT: a MATLAB and Python Toolbox for Optical Brain Mapping. Optical Tomography and Spectroscopy, Jan 2024.
- Ludwig DR, **Thacker Y**, et al. CT-derived textural analysis parameters discriminate high-attenuation renal cysts from solid renal neoplasms. Clinical Radiology, Jul 2023.
- Speh E, Segel A, **Thacker Y**, et al. NeuroDOTpy: A Python Neuroimaging Toolbox for DOT. Bio-Optics: Design and Application, Jan 2023.
- Speh E, **Thacker Y**, et al. NeuroDOT: A Python Neuroimaging Toolbox for DOT. Clinical and Translational Biophotonics, Jan 2022.
- Thacker Y**, et al. Integrating CLARA Workflow in XNAT as an Image Processing Pipeline; Predicting Mortality of COVID-19 Positive Patients in the ICU. XNAT Workshop (Posters), Sep 2021.

TECHNICAL SKILLS AND COURSEWORK

- Languages and Technologies:** Python, MATLAB, Docker, Bash, Java, C#, JavaScript, HTML, NVIDIA Clara, TensorFlow, Keras, **scikit-learn**, NumPy, Pandas, Matplotlib, SLURM, XNAT, BIDS, DICOM, FSL, FreeSurfer.
- Imaging and ML Methods:** Transport-Based Morphometry, diffusion MRI, tractography, connectomics, fNIRS/DOT, radiomics, CNNs, PCA, LDA, Random Forests, GLM regression, FDR correction, cross-validation, confound regression.
- Relevant Coursework:** Algorithms in Computational Biology, Matrix Algebra, Parallel and Concurrent Programming, Object-Oriented Programming, Data Structures and Algorithms, Anatomy and Physiology, Genetics, Molecular Biology.
- Laboratory Techniques:** Mini-preps, In-Fusion cloning, restriction digests, gel electrophoresis, PCR, SDS-PAGE.